Introduction to Linear Regression

Revision sheet — Data Analysis for Psychology in R 2 (DAPR2), Week 1 · University of Edinburgh 2026–27

Learning objectives (= what can be examined)

- Understand the link between **models** and **functions**.
- Understand the key concepts of the linear model: **intercept** and **slope**.
- Understand what **residuals** represent.
- Understand the key principles of **least squares**.
- Be able to specify a simple linear model in R with `lm()`.

1. Models and functions

A **model** is a formal representation of a system — put another way, *an idea about the way the world is*. Pretty much all of statistics is about building and testing models.

We represent mathematical models as **functions**: expressions that define the relationship between one variable (or set of variables) and another. A function specifies *what is important* (the arguments) and *how they combine* (the operations). This is what lets us **make and test predictions**.

Lecture example (babies). Length (cm) = 55 + 4 × Age (months). At 10 months the model predicts 95 cm; at 11 months, 99 cm. Predictions are points that fall **on the line**. But at age 20 (240 months) it predicts 1,015 cm — over 10 metres: absurd. Real data show that growth is **not linear** across the whole age range. **Lesson:** we judge a model by comparing its predictions with observed data; if they deviate, the model is not good (and beware of **extrapolating** beyond the observed range).

Deterministic model	**Statistical model**
An exact relationship: y = 3 + 2x	Allows case-by-case variability: y = 3 + 2x + ε
Every point lies exactly on the line.	Points scatter around the line; ε captures the deviation.

2. The linear model

DAPR2 focuses on the **linear model**: it assumes the relationship between the **outcome** variable and the **predictor(s)** is linear, and describes the outcome as a function of one or more predictors. Blocks 1–3: continuous numeric outcomes; Block 4: binary outcomes (logistic regression).

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

Symbol	Name	Meaning (example: hours studied → test score)
y_i	outcome (DV)	Test score of participant i
x_i	predictor (IV)	Hours studied by participant i
β_0	intercept	Value of y when x = 0 (height of the line)
β_1	slope	Change in y for a 1-unit increase in x (gradient)
ε_i	residual / error	Difference between observed and predicted value for i

Why is there an i on some terms but not others?

Classic exam question

- **i** is a subscript indicating that **each participant has their own value**: their own score y_i, hours x_i and residual ε_i.
- β_0 and β_1 have **no subscript**: there is **one value for all observations**. The model (the line) is for all of our data.

3. Interpreting the intercept and slope

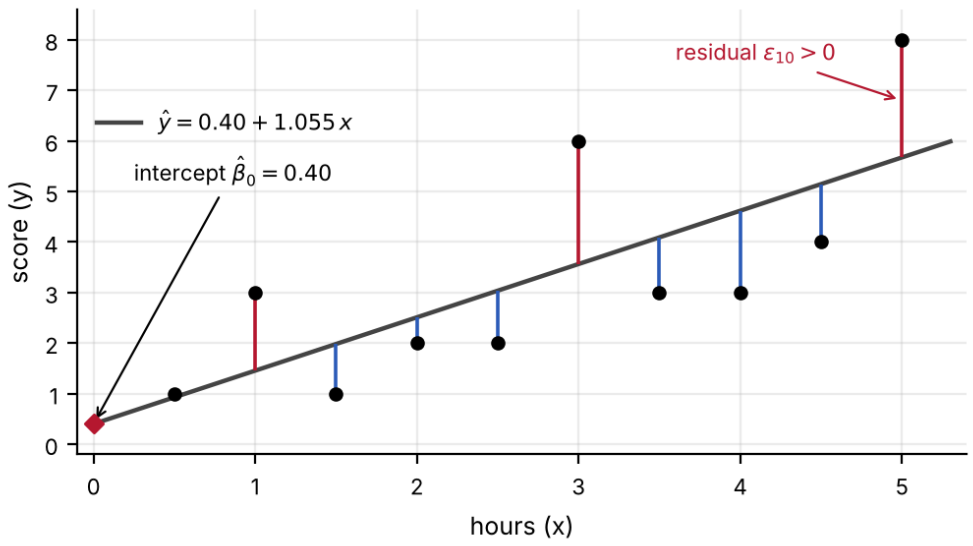
- **Intercept (β_0)**: where the line crosses ("intercepts") the y-axis, i.e. the predicted value of y when **x = 0**. E.g. intercept = −5 → the line passes through (0, −5).
- **Slope (β_1)**: the gradient, or rate of change. When x increases by **1 unit**, predicted y changes by β_1. E.g. slope = −2 → each unit of x lowers y by 2. Positive slope = line goes up, negative = goes down, 0 = flat.
- These two values are the **parameters** (or **coefficients**) of the model: together they fully define the line.

4. Residuals

The residual ε_i is the **vertical distance** between the observed data point and the model-predicted line:

$$\varepsilon_i = y_i - \hat{y}_i \quad \text{(observed − predicted)}$$

- $\varepsilon_i > 0$: the point is **above** the line (the model under-predicts) — red below.
- $\varepsilon_i < 0$: the point is **below** the line (the model over-predicts) — blue below.
- Residual = error: the smaller the residuals overall, the better the model fits the data.



Lecture data (10 students) with the least-squares line and residuals (vertical segments).

5. The principle of least squares

The true parameters β_0, β_1 are **unknown** and must be **estimated** from the data. The best estimates are written with a hat, β-hat (here shown as $\mathbf{b_0}$, $\mathbf{b_1}$, the notation the lecture uses for R). A good line is one that is **"close" to all the points** — in the least-squares sense.

The 4-step procedure

1) Define some line → 2) Calculate the residuals → 3) Square them → 4) Sum the squares. **The best line is the one that minimises the residual sum of squares.**

$$SS_{Residual} = \Sigma_{i=1}^{n} (y_i - \hat{y}_i)^2 \quad \text{where} \quad \hat{y}_i = b_0 + b_1 x_i$$

Term	Meaning
y_i (data)	What we measured in our study (here: test scores)
$\hat{y}_i$ (predicted value)	What the model predicts given someone's predictor values (e.g. "if I study for 4 hours, what score?")
$y_i - \hat{y}_i$ (residual)	The difference between the two

"Why do we square the residuals?" (asked in the lecture)

- Without squaring, positive and negative residuals would **cancel out** (for the least-squares line they sum to exactly 0).
- Squaring makes every deviation **positive** and **penalises large deviations more**.
- A sum of squares is smooth to minimise, so there is a **closed-form solution** (formulas below).

Key point

The values of the intercept and slope that **minimise $\mathbf{SS_{Residual}}$** are **our estimated coefficients**. Minimising $SS_{Residual}$ means that, across all our data, the predicted values are **as close as they can be** to the observed values of the outcome.

6. Calculation formulas

Slope	Intercept
$\mathbf{b_1 = SP_{xy} / SS_x}$ $SP_{xy} = \Sigma(x_i - \bar{x})(y_i - \bar{y})$ (sum of cross-products) $SS_x = \Sigma(x_i - \bar{x})^2$ (sum of squares of x)	$\mathbf{b_0 = \bar{y} - b_1 \bar{x}}$ $\bar{x}$, $\bar{y}$ = means of x and y. Consequence: the line always passes through the point of means ($\bar{x}$, $\bar{y}$).

Worked example (lecture data)

hours x	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
score y	1	3	1	2	2	6	3	3	4	8
$\hat{y}$	0.93	1.45	1.98	2.51	3.04	3.56	4.09	4.62	5.15	5.67
residual	+0.07	+1.55	-0.98	-0.51	-1.04	+2.44	-1.09	-1.62	-1.15	+2.33

$\bar{x} = 2.75$; $\bar{y} = 3.3$; $SS_x = 20.625$; $SP_{xy} = 21.75$ ⇒ $\mathbf{b_1 = 21.75 / 20.625 = 1.0545}$; $\mathbf{b_0 = 3.3 - 1.0545 \times 2.75 = 0.40}$. $SS_{Residual} = 21.16$. Prediction for 4 hours: $0.40 + 1.0545 \times 4 \approx \mathbf{4.62}$.

7. Simple linear model in R

$\hat{y}_i = b_0 \cdot 1 + b_1 \cdot x_i$ → maps directly onto R formula notation `y ~ 1 + x` (the 1 is the intercept, included by default, so `y ~ x` is equivalent). The b's are the **coefficients** (estimated by the model); 1 and x_i are the **inputs** we provide.

- Syntax: `lm(DV ~ IV, data = datasetName)`
- Outcome (DV) on the **left** of ~, predictor(s) (IV) on the **right**.
- The dataset must contain columns with **exactly** the names used in the formula.

```
test <- tibble(
  student = paste(rep("ID", 10), 1:10, sep = ""),
  hours   = seq(0.5, 5, .5),
  score   = c(1, 3, 1, 2, 2, 6, 3, 3, 4, 8)
)
mod1 <- lm(score ~ hours, data = test)
mod1            # coefficients: (Intercept) 0.400   hours 1.055
summary(mod1)   # detailed output
```

Reading the `summary(mod1)` output

```
Residuals:
    Min      1Q  Median      3Q     Max
-1.6182 -1.0773 -0.7454  1.1773  2.4364
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.4000     1.1111   0.360   0.7282
hours         1.0545     0.3581   2.945   0.0186 *
Residual standard error: 1.626 on 8 degrees of freedom
```

Line / column	What you should be able to say
Residuals	Distribution of the residuals (min, quartiles, median, max).
(Intercept) Estimate = 0.40	Predicted score for a student who studied **0 hours**: 0.40.
hours Estimate = 1.0545	Each additional hour of study is associated with a **1.05-point** increase in predicted score.
Std. Error, t value, Pr(>|t|)	Uncertainty and significance test for each coefficient (covered in later weeks). Here the slope is significant (p = .019 < .05), the intercept is not (p = .73).
Residual standard error = 1.626	Spread of the residuals; df = n − 2 = 10 − 2 = **8** (two parameters estimated).

Must remember

- A **model** describes how variables relate; a **function** formalises it and generates **predictions**.
- The linear model describes data with an **intercept** and slope(s): $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$.
- **Residual** = error = observed − predicted: tells us how good (or bad) the model is.
- Coefficients are found by **least squares**: they minimise the sum of squared residuals.
- $b_1 = SP_{xy}/SS_x$; $b_0 = \bar{y} - b_1\bar{x}$; in R: `lm(y ~ x, data = d)`.

Common pitfalls

- Confusing **ε** (residual, one per participant) with **β** (one parameter for everyone).
- Flipping the residual's sign: it is **observed − predicted**, not the reverse.
- Reading the intercept as a mean: it is the prediction at **x = 0** (which may be meaningless in context).
- Extrapolating far beyond the data (the 10-metre baby!).
- Putting the DV on the right of ~ in `lm()`.